

Member ID: _____

Time: _____

Rank: _____



COMPUTER SECURITY (320)

NATIONAL 2022

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each)

_____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice: Identify the letter of the choice that *best* completes the statement or answers the question.

- 1) What is the best practice when working with accounts of terminated users?
 - a) delete the account
 - b) change the password
 - c) do nothing
 - d) disable the account
- 2) What is the process of giving individual access to a system or resource?
 - a) authentication
 - b) authorization
 - c) accounting
 - d) auditing
- 3) Which of the following is not a profile in Windows firewall?
 - a) public
 - b) domain
 - c) workgroup
 - d) private
- 4) If ping fails, what firewall rule might be blocking it?
 - a) Core Networking (Echo Request – ICMPv4-In)
 - b) File and Printer Sharing (Echo Request – ICMPv4-In)
 - c) Core Networking – IPv6 (IPv6-In)
 - d) File and Printer Sharing (LLMNR-UDP-In)
- 5) What program should be used to manage firewall rules in Windows?
 - a) Malware Bytes Anti-Malware
 - b) Windows Firewall
 - c) Security Center
 - d) Windows Defender Firewall with Advanced Security
- 6) What is the best practice for using the built-in administrator account in Windows?
 - a) Rename it
 - b) Delete it
 - c) Use it
 - d) Disable it

- 7) What is the most common form of authentication?
 - a) password
 - b) pin
 - c) digital certificate
 - d) smart cards

- 8) What type of attack intercepts communication between parties to steal or manipulate the data?
 - a) replay
 - b) MAC spoofing
 - c) man-in-the-middle
 - d) ARP poisoning

- 9) What seven-layer model is often used to describe networking technologies and services?
 - a) TCP/IP
 - b) OSI
 - c) DIX
 - d) IPX/SPX

- 10) What technology built-in to Windows allows users to encrypt their files and volumes?
 - a) RSA
 - b) File Explorer
 - c) CryptKey
 - d) Bitlocker

- 11) Which of the following is not a characteristic of a complex password?
 - a) 5 or more characters
 - b) lowercase
 - c) numbers
 - d) special characters

- 12) What technology is used to scan a fingerprint for authentication?
 - a) three-factor authentication
 - b) scanner
 - c) biometrics
 - d) smart card

- 13) What character can be used to create a hidden file share?
 - a) @
 - b) \$
 - c) %
 - d) *

- 14) What type of electronic document contains a public key?
- a) digital certificate
 - b) biometrics
 - c) PIN
 - d) PAN
- 15) What do you call multiple Windows updates that have been packaged together as one installation and are well tested?
- a) service packs
 - b) cumulative packs
 - c) critical update
 - d) optional update
- 16) In which type of encryption is the same key used to encrypt and decrypt data?
- a) public
 - b) private
 - c) symmetric
 - d) asymmetric
- 17) At what layer of the OSI model does the IP protocol function?
- a) Transport Layer
 - b) Network Layer
 - c) Data Link Layer
 - d) Presentation Layer
- 18) What protocol can be used by a host on a network to find the MAC address of another device based on an IP address?
- a) DNS
 - b) ARP
 - c) TCP
 - d) UDP
- 19) What type of device looks at packets and forwards it based on its destination IP address?
- a) bridge
 - b) switch
 - c) router
 - d) hub
- 20) What item, about the size of a credit card, allows access to a network and its resources?
- a) digital certificate
 - b) smart card
 - c) security token
 - d) PIN

- 21) What term describes taking advantage of someone's natural tendencies and emotional responses in order to gain access to data?
- a) phreaking
 - b) hacking
 - c) social engineering
 - d) reverse engineering
- 22) What type of malware can copy itself and infect a computer without the user's consent or knowledge?
- a) virus
 - b) Trojan horse
 - c) rootkit
 - d) backdoor
- 23) A _____ is a device that can monitor a network passively.
- a) Honeypot
 - b) IDS
 - c) Sniffer
 - d) Trapdoor
- 24) What malware looks like a useful or desirable program but is in reality a program that is supposed to cause harm to your computer or steal information from your computer?
- a) virus
 - b) Trojan horse
 - c) worm
 - d) backdoor
- 25) What is a honeypot?
- a) A host on the network that is meant to be broken into by another attacker
 - b) A host on the network that manages all the storage devices
 - c) A network device that monitors the flow of traffic into the network
 - d) A switch that has been modified to relay intercepted information to a third party
- 26) What do you call the security discipline that requires that a user is given no more privilege than is necessary to perform their job?
- a) defense in depth
 - b) reduction of attack surface
 - c) risk transfer
 - d) principle of least privilege

- 27) What malware collects a user's personal information or details about their browsing habits without their knowledge?
- a) virus
 - b) Trojan horse
 - c) worm
 - d) spyware
- 28) What type of policy can be configured to protect against users trying to brute force a password?
- a) auditing policy
 - b) password policy
 - c) account lockout policy
 - d) security policy
- 29) What is the Windows utility that can be used to set password requirements on an individual computer in a workgroup?
- a) Local Security Policy
 - b) Local Group Policy Manager
 - c) Windows Defender
 - d) Users and Groups
- 30) What is the best way to protect against social engineering?
- a) stronger encryption
 - b) stronger authentication
 - c) employee awareness
 - d) risk mitigation
- 31) _____ is the use of fraudulent emails pretending to be a reputable company in order to induce individuals into providing their personal information, such as passwords and credit card numbers.
- a) phishing
 - b) social engineering
 - c) hacking
 - d) key logging
- 32) What is the first line of defense when setting up a network?
- a) physically securing the network
 - b) configure authentication
 - c) configure encryption
 - d) configure an ACL

- 33) Phone numbers, addresses, and credit card numbers are all examples of _____.
a) private data
b) PPI
c) things that should be encrypted
d) info you keep in an address book
- 34) Which is a best practice for password security?
a) writing it in a journal in your desk drawer
b) using the same password everywhere so you can easily remember it
c) changing your password every 90 days
d) using personal information to make it memorable
- 35) What term can be described as a function of threats, consequences of those threats, and the resulting vulnerabilities?
a) threat
b) mitigation
c) risk
d) management
- 36) Which computer Security Word best describes checking your driver's license to verify you are who you say?
a) Authenticated
b) Authorized
c) Login Credential
d) Password
- 37) What program should be used to manage firewall rules in Windows?
a) Malware Bytes Anti-Malware
b) Windows Firewall
c) Security Center
d) Windows Defender Firewall with Advanced Security
- 38) If ping fails to your machine, you need to create a rule in windows firewall. Which one?
a) Allow - (Echo Request – ICMPv4-In)
b) Enable File and Printer Sharing (Echo Reply ICMPv4-in)
c) Core Networking – IPv6 (IPv6-In)
d) File and Printer Sharing (LLMNR-UDP-In)
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 - b) hacking
 - c) social engineering
 - d) reverse engineering
- 50) What type of malware can copy itself and infect a computer on every reboot, without the user's consent or knowledge?
- a) virus
 - b) Trojan horse
 - c) rootkit
 - d) backdoor